

MULTI-AGENT URBAN TRAFFIC SIGNAL OPTIMIZATION IN SIMULATED ENVIRONMENTS

Project Id: 25-26J-523

Project Proposal Report

Randika D. G. T.

B.Sc. (Hons) Degree in Information Technology Specialized in Software
Engineering

Department of Computer Science and Software Engineering

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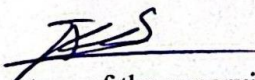
August 2025

DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
Randika D. G. T.	IT22598714	

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.


Signature of the supervisor:

Date: 20-08-25 .

ABSTRACT

Urban traffic congestion significantly impedes emergency vehicle (EV) response times, a problem exacerbated by the reactive and often unsafe nature of traditional preemption systems. A critical research gap exists in the development of holistic systems that integrate proactive control, intelligent safety analysis, and real-time driver feedback. This project addresses this gap by proposing a comprehensive, AI-driven Emergency Vehicle Preemption System (EVPS) designed to operate within a high-fidelity SUMO simulation. The system's architecture is built on a multi-layered design that includes a predictive ETA model using a Long Short-Term Memory (LSTM) network for proactive signal control, and an interpretable Decision Tree model for real-time conflict minimization to ensure safety. Furthermore, the system closes the human-in-the-loop communication gap by providing real-time visual feedback to the EV driver via a mobile application. This integrated approach aims to significantly reduce EV travel times while minimizing disruption and enhancing the overall safety of preemption maneuvers, providing a robust and verifiable solution to a persistent challenge in intelligent transportation systems.

Keywords: Emergency Vehicle Preemption, Intelligent Transportation Systems, SUMO, LSTM, Decision Tree, ETA Prediction, Traffic Simulation.

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LIST OF ABBRIVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ETA	Estimated Time of Arrival
EV	Emergency Vehicle
EVPS	Emergency Vehicle Preemption System
GNN	Graph Neural Network
IoT	Internet of Things
ITS	Intelligent Transportation Systems
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MPC	Model Predictive Control
RMSE	Root Mean Square Error
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
UI	User Interface
V2I	Vehicle-to-Infrastructure
VANET	Vehicular Ad-hoc Network

1. INTRODUCTION

Urban traffic congestion is a critical and persistent challenge for modern cities, leading to significant economic losses, increased travel times, and severe environmental impacts [1]. In many urban centers, particularly in developing regions like Sri Lanka, traffic signal infrastructure relies on fixed-time or basic actuated logic. These systems are inherently unresponsive to the dynamic, real-time fluctuations in traffic volume and cannot adapt effectively to non-recurring events such as accidents or road closures [2]. This inflexibility is a primary cause of network-wide inefficiency and poses a direct threat to public safety by impeding the rapid response of emergency services.

A critical function of urban traffic management is ensuring the swift passage of emergency vehicles (EVs). However, the effectiveness of traditional Emergency Vehicle Preemption (EVP) systems is often limited. Most existing systems are reactive, triggering a signal override only when an EV is already in close proximity to an intersection [3]. This last-minute approach forces abrupt signal changes, fails to account for downstream traffic conditions, and can be inefficient in clearing existing vehicle queues.

Furthermore, a significant deficiency identified across the literature is the lack of integrated safety analysis in preemption logic. The primary goal is often to provide a green light for the EV, but this is pursued without a holistic assessment of the intersection's state, potentially creating hazardous secondary effects and increasing the risk of collisions [4]. This myopic focus on the EV's path neglects the complex dynamics of the surrounding traffic network. While modern research has begun to address these issues by leveraging AI with studies proposing the use of Deep Reinforcement Learning for post-preemption traffic recovery [5] and mixed-criticality systems for managing multiple EVs [4] the core limitations of reactive control and inadequate safety considerations persist.

This research directly addresses these gaps by proposing a proactive, safety-aware Emergency Vehicle Preemption System (EVPS). The system is designed to operate within a larger multi-agent simulation, tackling the advanced research question of how to manage high-priority preemption when the background traffic control system is itself adaptive [6].

By integrating predictive AI models for proactive control, a dedicated classifier for intelligent safety analysis, and a human-in-the-loop interface for driver feedback, this project aims to develop a comprehensive solution that minimizes EV travel time while ensuring safety and mitigating disruption to the overall traffic network.

1.1 Background and Literature Review

Emergency Vehicle Preemption (EVP) is a critical component of Intelligent Transportation Systems (ITS), designed to grant right-of-way to emergency vehicles (EVs) at signalized intersections. The primary goal of any EVP system is to reduce the travel time for first responders, which is a key factor in improving outcomes in life-threatening situations. Traditional EVP systems often rely on simple, line-of-sight triggers or GPS-based geofencing, which, while functional, are fundamentally reactive. These reactive systems often lead to abrupt signal changes and do not efficiently manage the subsequent traffic disruption [3]. Modern research has therefore shifted towards proactive, intelligent systems that leverage advancements in AI and communication technologies to create safer and more efficient preemption.

A cornerstone of a proactive EVP system is the ability to accurately predict an EV's Estimated Time of Arrival (ETA) at upcoming intersections. This predictive capability allows the system to prepare intersections in advance, creating a smooth "green wave" rather than forcing a last-second signal change. Recent literature strongly indicates that Long Short-Term Memory (LSTM) networks are exceptionally well-suited for this task due to their inherent ability to model complex, time-dependent patterns in traffic data. For instance, the effectiveness of using an LSTM model to analyze historical and real-time data to forecast congestion has been demonstrated, achieving high accuracy and outperforming other models [7]. Furthermore, the challenge of data scarcity for emergency vehicles—a significant hurdle in training reliable models—has been addressed by frameworks like TLETA, which uses deep transfer learning to adapt models trained on data-rich regular vehicles to the EV domain with robust results [8]. This body of work

confirms that an LSTM-based predictor is a state-of-the-art foundation for an intelligent preemption system.

Beyond efficiency, a key limitation in many existing EVP systems is the lack of integrated safety analysis. The singular focus on providing a green light for an EV without considering the state of conflicting traffic can create hazardous conditions [4]. This highlights the need for a dedicated safety-aware component. For such a safety-critical task, the model must be not only accurate but also interpretable and verifiable. Research in the adjacent field of autonomous driving provides a strong justification for using tree-based models for this purpose, as Decision Trees have been shown to offer a compelling balance of real-time performance and accuracy while being inherently human-interpretable and formally verifiable [9]. This allows for a clear understanding of the model's decision-making process and the ability to mathematically prove its behavior under specific conditions, a level of assurance that is crucial for a safety module like a Conflict Minimization Classifier.

Finally, the operational context of modern traffic systems is increasingly dynamic and interconnected. Research has identified the need for EVP systems that can integrate with adaptive background traffic controllers and provide feedback to the EV driver [6]. Specifically, the development of "in-vehicle communication infrastructure" is a recognized need to provide drivers with real-time information, closing the human-in-the-loop gap [3]. Recent advancements in this area include V2X-enabled alert systems that use a dedicated driver interface, such as a mobile application, to display real-time warnings about route blockages and collision risks, directly enhancing the driver's situational awareness [10]. An effective, modern EVP system must therefore integrate these concepts, being not only predictive and safe but also providing clear, actionable feedback to the end-user.

1.2 Research Gap

A thorough review of the current literature reveals that while significant progress has been made in Emergency Vehicle Preemption, existing systems often operate as isolated solutions, failing to provide a holistic, integrated framework. This creates several critical research gaps that this project aims to address directly.

First, a predominant gap is the lack of integrated, intelligent safety analysis. Many systems, such as post-preemption recovery strategies, focus on mitigating traffic disruption but do not include a proactive mechanism to assess the safety of the preemption maneuver itself [5]. Although some research addresses multi-EV scenarios, it is also noted that most systems prioritize an EV's travel time without a robust method for arbitrating vehicular conflicts at the intersection [4]. This leaves a clear need for a dedicated component that can analyze the real-time traffic state to ensure a signal override is not only efficient but also safe.

Second, while research has explored standalone alert systems to improve driver situational awareness through in-vehicle interfaces [10], these systems reveal a more nuanced research gap: they are often not integrated with the automated signal preemption logic. This creates a critical disconnect where the driver may receive alerts about environmental hazards but lacks confirmation that the traffic infrastructure is actively clearing a path for them. The absence of this integrated feedback loop means the human element—the driver who must ultimately navigate the intersection—is still operating with incomplete information, a gap often neglected in system design [3].

Finally, many proposed solutions do not fully integrate the core components of a modern EVP system—predictive control, safety analysis, and driver feedback—into a single, cohesive architecture. While some studies focus on advanced routing [6] and others on post-event recovery [5], there is a lack of research demonstrating a system that combines a predictive ETA model for proactive control, an intelligent safety classifier, and a real-time driver interface. This project directly fills this void by developing a multi-layered system where each component addresses a specific limitation identified in the literature.

The following table provides a comparative analysis of the proposed system against key research, illustrating how it uniquely addresses these identified gaps.

Table 1.1 Comparison of Previous Researches

Features	Humagain & Sinha [4]	Khoirunnisaa et al. [5]	Chen et al. [6]	Kumar et al. [10]	Proposed System
Predictive ETA Model	×	×	✓	×	✓
Intelligent Safety / Conflict Analysis	×	×	×	✓	✓
Real-Time Driver Feedback Interface	×	×	×	✓	✓
Dynamic Route Adaptability	×	×	✓	×	✓

1.3 Research Problem

The central research problem is the design and implementation of a holistic, AI-driven Emergency Vehicle Preemption System (EVPS) that addresses the critical, interconnected challenges of **proactivity, safety, and driver communication**, which are inadequately handled by current systems. While existing solutions often focus on singular aspects of preemption, they fail to provide an integrated framework that solves the problem in its entirety.

The specific challenges this research aims to address are:

1. **How to transition from reactive to proactive signal control?** Most current EVP systems are reactive, triggering a signal override only when an EV is in immediate proximity. This approach is inefficient and disruptive. The problem lies in developing a predictive system that can accurately forecast an EV's arrival

time well in advance, enabling it to proactively and smoothly prepare a "green wave" rather than forcing an abrupt, last-minute intervention.

2. **How to ensure preemption is safe, not just fast?** A significant flaw in many preemption strategies is the myopic focus on granting a green light to the EV without a comprehensive safety assessment. This can create hazardous secondary effects by disrupting established traffic flows and increasing the risk of collisions on conflicting lanes. The research problem is to design an intelligent decision-making module that can perform a real-time safety analysis of the intersection's state and determine if a preemption is safe to execute at any given moment.
3. **How to close the human-in-the-loop communication gap?** EV drivers currently operate with significant uncertainty, lacking confirmation that the path ahead has been cleared. This information gap poses a safety risk and adds to the cognitive load on the driver. The problem is to develop a robust communication channel that provides the driver with clear, real-time, and actionable feedback, confirming the status of upcoming signals and transforming the preemption system from a "blind" operation into an interactive, human-in-the-loop process.

This project will address these questions by creating a single, cohesive system that integrates predictive modeling, intelligent safety classification, and a real-time driver interface to deliver a comprehensive solution to the multifaceted challenge of modern emergency vehicle preemption.

2. OBJECTIVES

2.1 Main Objectives

The main objective of this research is to design, implement, and evaluate a holistic, AI-driven Emergency Vehicle Preemption System (EVPS). This system aims to proactively reduce emergency vehicle travel times by creating intelligent "green waves" while ensuring the safety of the maneuver and providing real-time, in-vehicle feedback to the EV driver within a high-fidelity SUMO simulation environment.

2.2 Specific Objectives

To achieve the main objective, the following specific, measurable, and time-bound objectives will be pursued:

To develop and evaluate a predictive Estimated Time of Arrival (ETA) model for emergency vehicles.

This will be achieved by designing and training a Long Short-Term Memory (LSTM) network on synthetic trajectory data generated from SUMO. The model's performance will be evaluated based on its accuracy in forecasting an EV's arrival time in seconds to the next intersection.

To implement and validate an intelligent Conflict Minimization Classifier for safe preemption.

This objective involves training an interpretable, tree-based model (Decision Tree) to perform real-time safety assessments. The model will classify an intersection's state as either "Safe to Preempt" or "Unsafe to Preempt" based on vehicular traffic data, and its effectiveness will be validated against a rule-based heuristic.

To design and build a dynamic Green-Wave Controller to orchestrate signal preemption.

This involves developing the core logic that integrates the ETA predictor and the conflict classifier. The controller will use the predicted ETA to trigger the safety check and, upon receiving a "Safe" classification, will execute the signal override commands (all-red phase followed by green phase) via the SUMO Traci API.

To create and integrate a human-in-the-loop mobile application for EV driver feedback.

This objective will be met by developing a mobile user interface that connects to the simulation via a WebSocket. The application will provide the driver with real-time visual confirmation of the cleared route, the status of upcoming traffic signals, and any critical system alerts, thereby closing the communication gap.

3. METHODOLOGY

This research will follow a structured, iterative methodology centered on system design, synthetic data generation, AI model development, and empirical evaluation, all conducted within the SUMO (Simulation of Urban Mobility) simulation platform. The proposed system operates within a larger multi-agent traffic optimization framework, the overall architecture of which is depicted in Figure 3.1. The AI-driven Emergency Vehicle Preemption System (EVPS) is designed with a multi-layered architecture to ensure modularity and a clear separation of concerns, as detailed below.

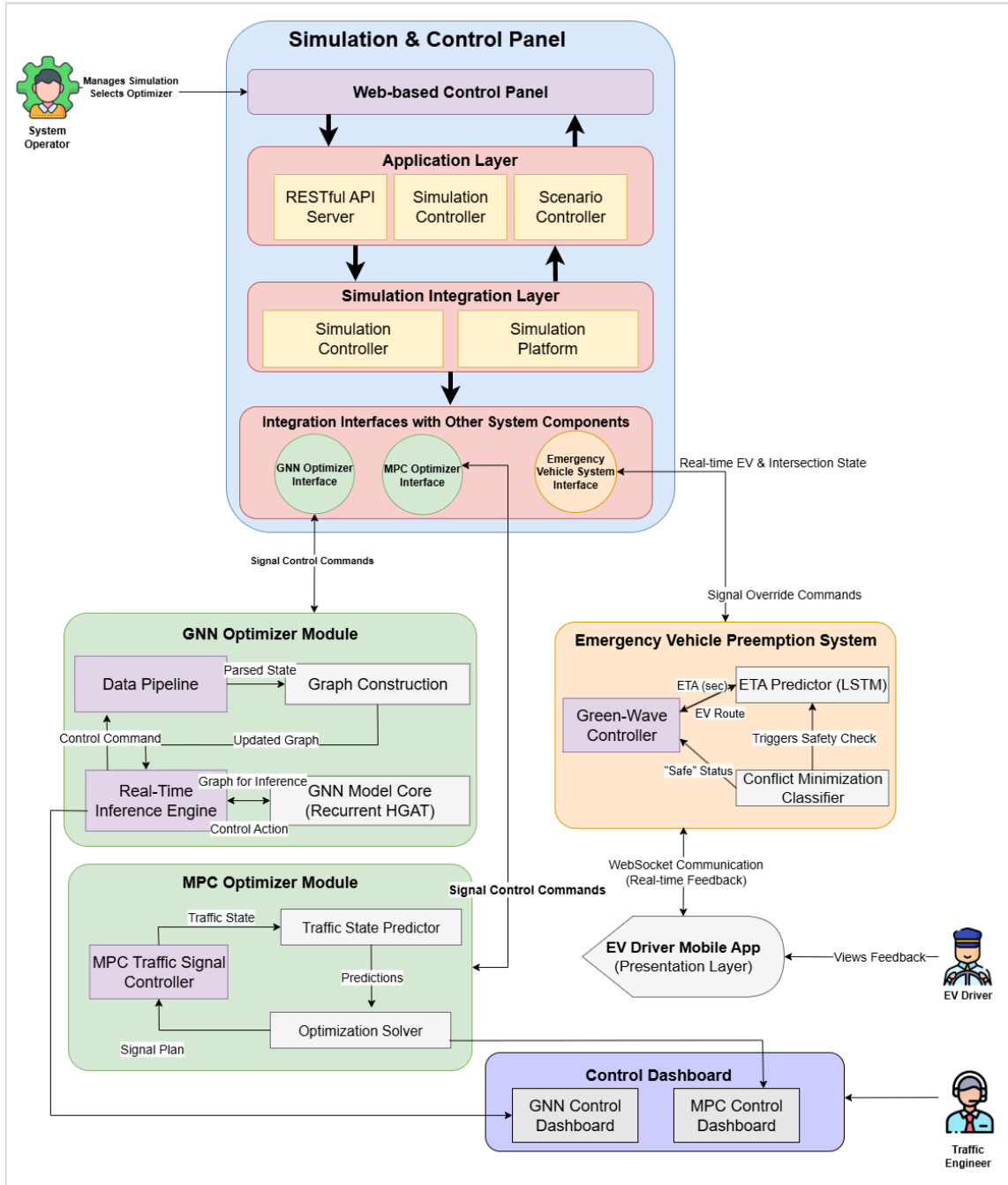


Figure 3.1 System diagram for the overall system

The architecture of the EVPS component, illustrated in Figure 3.2, is composed of four primary layers that work in concert. The Data Interface Layer serves as the crucial bridge to the SUMO simulation, implemented using the Traci API to subscribe to real-time EV data and execute signal override commands. At the core of the system is the Decision & Control Layer, which houses the main AI logic. This layer includes an LSTM-based ETA Predictor for forecasting arrival times, a Decision Tree-based Conflict Minimization Classifier for safety checks, and a Green-Wave Controller that orchestrates the preemption process. Finally, the Presentation Layer provides the human-in-the-loop feedback through a mobile application, which connects via a WebSocket to display real-time route and signal status information to the EV driver.

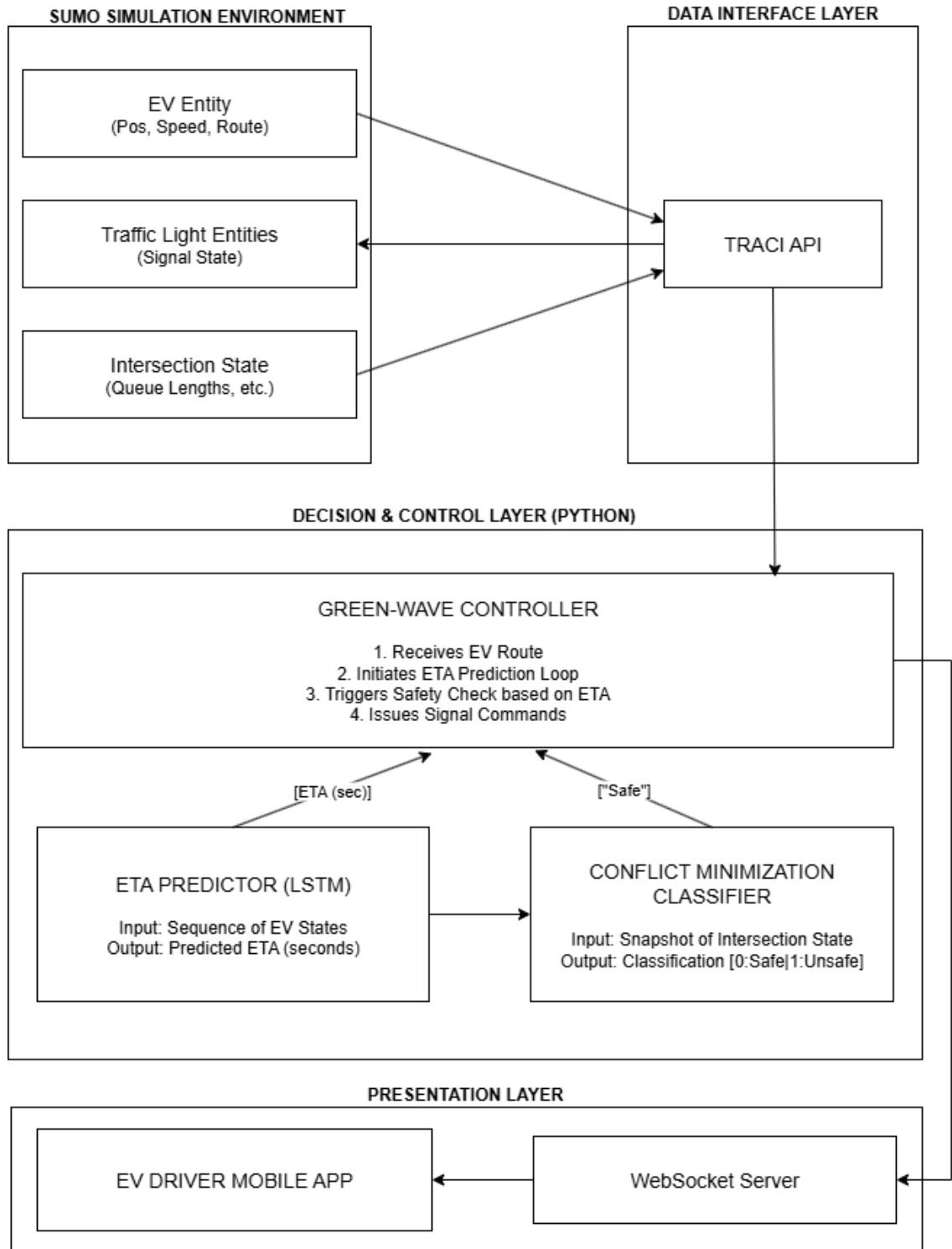


Figure 3.2 System architecture diagram for the component

The development of the AI models will utilize distinct architectures tailored to their specific tasks. To capture the time-series nature of vehicle movement, the ETA Predictor will be an LSTM network implemented in TensorFlow/Keras. For the safety-critical task of conflict minimization, an interpretable Decision Tree model will be implemented using Scikit-learn to prioritize transparency and verifiability.

As real-world EV preemption datasets are not readily available, a synthetic data generation and training strategy will be employed. Thousands of diverse traffic scenarios will be scripted and run in SUMO to log vast amounts of training data via the Traci API. The data for the Conflict Minimization Classifier will be labeled using a predefined, rule-based heuristic that codifies safe and unsafe intersection states. While the models will be trained on this synthetic data, the SUMO simulation itself will be calibrated using real-world traffic flow data to ensure the generated patterns are realistic and the results are valid.

The performance of the completed EVPS will be rigorously evaluated using a combination of metrics. The primary success criterion will be the percentage reduction in average EV travel time compared to a baseline scenario without preemption. Secondary metrics will include the impact on non-EV traffic (measured by average delay and queue length), the classification accuracy of the Conflict Minimization Classifier, and the prediction error (MAE/RMSE) of the ETA Predictor. This comprehensive evaluation will be conducted across various traffic densities to assess the system's effectiveness and robustness.

4. PROJECT REQUIREMENTS

This section outlines the functional and non-functional requirements for the AI-driven Emergency Vehicle Preemption System (EVPS). These requirements define the system's capabilities and the quality attributes it must exhibit to be considered successful.

Functional Requirements

- FR1: EV Data Ingestion: The system shall connect to the SUMO simulation via the Traci API to continuously receive real-time data for a designated emergency vehicle, including its current position, speed, and pre-defined route.
- FR2: ETA Prediction: The system shall use the ingested time-series data to compute a predicted Estimated Time of Arrival (ETA) in seconds for the EV at its next signalized intersection.
- FR3: Conflict State Classification: The system shall be capable of querying the state of an intersection (e.g., queue lengths on conflicting lanes) and using an AI model to classify the current state as either "Safe to Preempt" or "Unsafe to Preempt."
- FR4: Signal Preemption Control: Upon receiving a "Safe to Preempt" classification, the system shall issue a sequence of commands to the SUMO simulation via the Traci API to override the target traffic signal. This sequence must include an all-red clearance phase followed by a green phase for the EV's direction of travel.
- FR5: Real-time Driver Feedback: The system shall transmit status updates regarding the preemption process to the driver's mobile application. This includes visual confirmation of the route, the status of upcoming signals, and system alerts.

Non-Functional Requirements

- NFR1: Performance & Low Latency: The end-to-end decision loop—from receiving simulation data to sending a signal command—must be completed with

minimal latency to be effective in a real-time context. The target for this entire process is under 500 milliseconds.

- NFR2: Predictive Accuracy: The AI models must meet defined performance thresholds.
- The ETA Predictor shall achieve a Mean Absolute Error (MAE) of less than 2 seconds on the test dataset.
- The Conflict Minimization Classifier shall achieve a classification accuracy of over 95% on the test dataset.
- NFR3: Interpretability: The decision-making logic of the Conflict Minimization Classifier must be human-readable. The final Decision Tree model should be visualizable to allow for verification of its learned rules.
- NFR4: Modularity: The system's architecture shall be modular, with clear separation between the data interface, the decision and control logic, and the presentation layer, to allow for independent development and testing.
- NFR5: Reliability: The WebSocket connection between the core system and the driver's mobile application must remain stable throughout a simulation run to ensure the driver receives continuous and uninterrupted feedback.

User requirements

- UR1: The EV driver must be able to view their assigned route and real-time position on a map-based interface within the mobile application.
- UR2: The EV driver must receive clear and unambiguous visual feedback on the status of upcoming traffic signals (e.g., a green icon for a cleared intersection, an amber icon for a pending clearance).
- UR3: The EV driver must be alerted via the mobile application to any system failures or critical issues, such as a failed preemption attempt.

Software requirements

- Simulation Environment: SUMO (Simulation of Urban Mobility)

- Programming Language: Python 3.8+
- Core Libraries:
 - Traci (for SUMO interaction)
 - TensorFlow / Keras (for the LSTM model)
 - Scikit-learn (for the Decision Tree model)
- Communication Protocol: WebSockets
- Mobile Development: A cross-platform framework such as Flutter or React Native

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